

AZ-900 Microsoft Azure Fundamentals – Complete Cheat Sheet & Exam Prep

Based strictly on the official Skills Measured outline

Use this guide for one-sitting review. It is written as a high-signal study packet, not as a full course.

Practice questions in this guide are original exam-style scenarios aligned to the official AZ-900 objectives and Microsoft Learn practice-assessment style. They are not leaked or memorized exam items.

Target audience	First-time Azure learners, non-technical stakeholders, junior IT staff, and anyone validating Azure fundamentals
Exam focus	Cloud concepts, Azure architecture/services, Azure management/governance
Best use	Final revision, lecture handout, quick-reference PDF, bootcamp recap

Prepared for rapid exam review and instructor-led delivery

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Exam overview

At a glance

Certification: Microsoft Certified: Azure Fundamentals (AZ-900)

Passing score: 700 / 1000

Exam time: 45 minutes exam time (typically 65 minutes total appointment time)

Typical question count: Microsoft says Fundamentals exams usually contain roughly 35-50 questions, but the exact number can vary

Delivery: Proctored exam with common certification question styles such as multiple choice, build-list/drag-and-drop style, and scenario questions

Important note: Microsoft Learn access is not available during Fundamentals exams

Tips for success

- Study by objective, not by service catalog. AZ-900 tests what Azure service or concept fits a scenario.
- Memorize comparisons: IaaS vs PaaS vs SaaS, VPN Gateway vs ExpressRoute, Azure Policy vs RBAC vs locks, Service Health vs Monitor vs Advisor.
- Learn the hierarchy cold: management groups -> subscriptions -> resource groups -> resources.
- Read questions for clue words such as *govern*, *deny*, *assign permissions*, *prevent deletion*, *estimate cost*, *monitor app performance*, and *private connection*.
- Do not overthink architecture depth. AZ-900 is breadth-first, not engineer-level implementation detail.
- Use the exam sandbox before test day so drag-and-drop/build-list wording does not slow you down.
- Answer every question. There is no penalty for guessing.

What AZ-900 is really testing

- Can you distinguish cloud concepts from Azure-specific services?
- Can you identify the best Azure service for a common business or technical need?
- Can you separate governance, security, identity, cost, deployment, and monitoring tools?
- Can you reason at the fundamentals level without needing deep implementation steps?

Describe cloud concepts (25-30%)

Describe cloud computing

What the exam tests

- Define cloud computing in plain language.
- Understand the shared responsibility model.
- Compare public, private, and hybrid cloud models and choose the right fit.
- Recognize the consumption-based model and common cloud pricing approaches.
- Identify what serverless means and when it fits.

Core definition

Cloud computing is the on-demand delivery of compute, storage, networking, databases, and software over the internet, typically billed by usage instead of up-front ownership.

Azure example: A company deploys a web app to Azure App Service instead of buying servers and managing the operating system.

Shared responsibility model

Responsibility shifts from customer -> provider as you move up the stack.

On-premises

Customer: Datacenter | Network | Servers | OS | Runtime | App | Data | Identity

IaaS

Microsoft: Datacenter | Network | Physical hosts

Customer : OS | Runtime | App | Data | Identity | Network controls

PaaS

Microsoft: Datacenter | Network | Physical hosts | OS | Runtime

Customer : App | Data | Identity | Configuration

SaaS

Microsoft: Almost everything in the stack

Customer : Data | Identity/access | Device/end-user settings

Exam tip: The customer is always responsible for their data and access decisions, even when Microsoft manages more of the stack.

Cloud models and use cases

Cloud model	Best description	Best fit use cases	Exam cue words
Public cloud	Services owned and operated by a cloud provider and delivered over the internet	Fast startup, elastic scale, global reach, low up-front cost	shared infrastructure, pay as you go, fastest deployment
Private cloud	Cloud environment dedicated to a single organization	Strict control, legacy constraints, highly specific compliance or isolation needs	dedicated, controlled, single-organization
Hybrid cloud	Mix of on-premises/private and public cloud with integration between them	Gradual migration, data residency, burst to cloud, keep some systems on-prem	connect on-prem to cloud, phased migration

Azure example: A bank keeps a sensitive legacy directory-integrated application on-premises but extends backup, monitoring, and some workloads into Azure. That is hybrid.

Consumption-based model and cloud pricing models

Model / concept	What it means	When exam answers point here
CapEx	Large up-front purchase of hardware/software	Traditional datacenter buying
OpEx	Ongoing operating expense based on usage	Core cloud financial model
Pay-as-you-go	Pay only for what you consume	Default Azure mental model
Reserved capacity / reservations / savings-style commitment	Lower price in exchange for committing to predictable use over time	Steady-state workloads
Spot pricing	Deep discounts for interruptible capacity	Fault-tolerant or batch jobs

Mnemonic: Cloud likes OpEx. Predictable use likes reservations. Interruptible work likes Spot.

Serverless

Serverless means you focus on code or workflows while Azure automatically handles infrastructure, scaling, and often per-execution billing.

Serverless trait	What to remember
No server management	You do not manage VM infrastructure
Event-driven	Runs on triggers such as HTTP requests, timers, queue messages
Automatic scaling	Scales out and in automatically
Consumption billing	Often pay for executions, run time, or events

Azure example: Azure Functions processes uploaded images only when files land in Blob Storage.

What is commonly tested

- “Pay only when code runs” -> serverless / Azure Functions.
- “No up-front purchase” -> OpEx / consumption-based model.
- “Dedicated for one organization” -> private cloud.
- “Keep some systems on-prem and some in cloud” -> hybrid cloud.

Describe the benefits of using cloud services

What the exam tests

- High availability and scalability
- Reliability and predictability
- Security and governance
- Manageability

Benefits summary

Benefit area	Meaning	Azure-flavored example	Typical exam wording
High availability	Service remains accessible despite failures	Deploying across availability zones	minimize downtime
Scalability	Increase or decrease resources to match demand	Scale VM count or App Service plan	scale up / scale out
Elasticity	Automatic or rapid scaling with demand changes	Autoscale during sales event	demand spikes, burst
Reliability	Ability to recover and continue operating	Replication, zones, paired regions	resilient, fault-tolerant
Predictability	Stable performance and cost planning using telemetry and tooling	Cost analysis, monitoring, autoscaling	forecast, plan, baseline
Security	Built-in controls and provider-managed infrastructure protections	Microsoft-managed datacenter security	secure by design
Governance	Policy-based standardization and compliance enforcement	Azure Policy, tags, management groups	compliant, standardize
Manageability	Manage via portal, CLI, templates, and automation	ARM templates and Azure portal	automate, centrally manage

Quick distinctions the exam likes

- **Scalability** = can grow or shrink.
- **Elasticity** = can automatically adjust with demand.
- **High availability** = minimize downtime.
- **Reliability** = system keeps working and can recover from failure.
- **Governance** = enforce standards.
- **Manageability** = deploy, monitor, and control efficiently.

Azure example: A retailer uses autoscale so its app adds compute during Black Friday and removes it overnight. That demonstrates scalability and elasticity.

Describe cloud service types

What the exam tests

- Compare IaaS, PaaS, and SaaS.
- Identify the right service type from a scenario.
- Recognize who manages what.

IaaS vs PaaS vs SaaS

Service type	You manage	Provider manages	Best use case	Azure / Microsoft examples
IaaS	OS, patches, app runtime, app, data, config	Datacenter, networking fabric, physical servers	Lift-and-shift, maximum flexibility, custom OS control	Azure Virtual Machines
PaaS	App code, data, app config	Infra, OS, runtime, scaling platform	Build/deploy apps fast without server admin	Azure App Service, Azure SQL Database, Azure Functions
SaaS	Data, users, access settings, limited app config	Entire application and platform	Consume ready-made software	Microsoft 365, Dynamics 365

Scenario cheat sheet

Scenario	Best answer
"I need full control of the operating system."	IaaS
"I want to deploy code without patching servers."	PaaS
"I just need business software users can sign into."	SaaS
"I am migrating an existing legacy server with minimal redesign."	IaaS
"I want Microsoft to manage the web hosting platform."	PaaS

Mnemonic: IaaS = rent infrastructure, PaaS = rent a platform, SaaS = use software.

Representative practice questions - Describe cloud concepts**Q1. Multiple choice**

A company runs a small image-processing API that receives unpredictable bursts of requests. It wants to avoid managing servers and pay only when requests arrive. Which option best fits?

- A. Azure Virtual Machines
- B. Azure Functions
- C. Azure Virtual Desktop
- D. Azure Managed Disks

Answer: B. Azure Functions

Why: Azure Functions is a serverless compute service designed for event-driven workloads with automatic scaling and consumption-based billing.

Skill tested: Describe cloud computing - serverless.

Q2. Scenario-based

An organization must keep a legacy manufacturing system on-premises, but wants to use Azure for backups, monitoring, and future migrations. Which cloud model is this?

- A. Public cloud
- B. Private cloud
- C. Hybrid cloud
- D. Community cloud

Answer: C. Hybrid cloud

Why: Hybrid cloud combines on-premises/private resources with public cloud resources and links them operationally.

Skill tested: Describe cloud computing - cloud models and appropriate use cases.

Q3. Multiple choice

Which statement best describes OpEx?

- A. Paying for datacenter hardware up front
- B. Paying ongoing costs as services are consumed
- C. Buying licenses once and using them forever
- D. Reserving capacity only for disaster recovery

Answer: B. Paying ongoing costs as services are consumed

Why: Cloud shifts many technology costs from capital expense (CapEx) to operating expense (OpEx).

Skill tested: Describe cloud computing - consumption-based model.

Q4. Drag-and-drop style

Match each scenario to the correct cloud service type.

Scenario	Correct match
Need full control over the OS for a migrated line-of-business application	IaaS
Need to deploy web code without managing the OS	PaaS
Need hosted email and collaboration software	SaaS

Why: The exam often tests service type selection by management responsibility and control level.

Skill tested: Describe cloud service types - IaaS, PaaS, SaaS.

Q5. Multiple choice

An application automatically adds resources when demand increases and removes them when demand drops. Which cloud benefit is most directly demonstrated?

- A. Governance
- B. Elasticity
- C. Private tenancy
- D. CapEx

Answer: B. Elasticity

Why: Elasticity is the ability to automatically adjust resources in response to changes in demand.

Skill tested: Describe the benefits of using cloud services - high availability and scalability.

Q6. Multiple choice

In a SaaS solution, which area always remains the customer's responsibility?

- A. Physical datacenter security
- B. Hypervisor patching
- C. Data and access decisions
- D. Host operating system maintenance

Answer: C. Data and access decisions

Why: Even in SaaS, the customer remains responsible for the data they store and how users access it.

Skill tested: Describe cloud computing - shared responsibility model.

Describe Azure architecture and services (35-40%)

Describe the core architectural components of Azure

What the exam tests

- Regions, region pairs, sovereign regions, availability zones, datacenters
- Resources and resource groups
- Subscriptions
- Management groups
- Azure hierarchy

Regions, datacenters, availability zones, and region pairs

Component	Meaning	Key exam point
Datacenter	Physical facility with power, cooling, and networking	Lowest physical layer
Azure region	Set of one or more datacenters in a geographic area	Deploy services close to users or to satisfy residency needs
Availability zone	Physically separate datacenter location within a region	Protect against datacenter-level failure
Region pair	A paired regional relationship used for resilience and recovery planning	Common exam cue for disaster recovery and update sequencing
Sovereign region	Isolated Azure instance for compliance, legal, or national requirements	Examples: Azure Government, Azure China

Geography

```
|
+-- Region A
|   +-- Availability Zone 1 (separate datacenter group)
|   +-- Availability Zone 2 (separate datacenter group)
|   +-- Availability Zone 3 (separate datacenter group)
|
+-- Region B (paired or alternate region for recovery)
```

Exam tips

- Availability zones are within a single region. They are for in-region resiliency.
- Region pairs relate to cross-region resiliency and update/recovery strategy.
- Not every new region has a classic pair; some Azure designs now rely on zonal architecture and region-level resiliency choices.

Azure example: Hosting a two-tier app across Availability Zone 1 and Availability Zone 2 in East US protects against a datacenter outage in that region.

Resources, resource groups, subscriptions, and management groups

Scope/component	What it is	What the exam wants you to know
Resource	Individual Azure item such as VM, storage account, or VNet	Basic deployable unit
Resource group	Logical container for resources	Resources belong to one resource group; used for lifecycle and management
Subscription	Boundary for billing, quotas, and access	Common separation for departments, environments, or billing

Scope/component	What it is	What the exam wants you to know
Management group	Governance scope above subscriptions	Apply policies/RBAC across multiple subscriptions

Important fundamentals

- A resource can belong to only one resource group.
- A resource group can contain resources in multiple regions.
- A subscription can contain multiple resource groups.
- Management groups help standardize governance across many subscriptions.

Azure hierarchy diagram

Management Group
 -> Subscription
 -> Resource Group
 -> Resource

Mnemonic: Govern high, deploy low.

Management groups govern subscriptions. Subscriptions contain resource groups. Resource groups contain resources.

Describe Azure compute and networking services

What the exam tests

- Compare compute types: containers, VMs, functions
- Compare VM options: Azure VMs, VM Scale Sets, availability sets, Azure Virtual Desktop
- Identify resources required for VMs
- Compare application hosting: web apps, containers, VMs
- Understand VNet, subnets, peering, Azure DNS, VPN Gateway, ExpressRoute
- Public endpoints vs private endpoints

Compare compute types

Compute type	Best for	Strengths	Limits / watch-outs
Virtual Machines	Maximum OS control, lift-and-shift workloads	Full admin control, custom software	You manage OS and more operations
Containers	Portable, fast-start apps and microservices	Consistent packaging, efficient deployment	Less OS-level control than VMs
Functions	Event-driven short tasks / serverless logic	No server management, scales automatically	Not ideal for long-running full server workloads

Azure example:

- VM -> migrated legacy app server
- Container -> packaged API running in Azure Container Apps or AKS
- Function -> process queue messages or webhook events

VM options comparison

Option	Best use	Key point the exam tests
Azure VMs	Single VMs or manually managed groups	Traditional IaaS compute

Option	Best use	Key point the exam tests
VM Scale Sets	Large sets of identical VMs with autoscaling	Scale horizontally automatically
Availability sets	Improve VM availability within a datacenter context	Separate fault/update domains for VMs
Azure Virtual Desktop	Centralized desktop/app delivery	Virtual desktops/apps from Azure

Quick distinction

- **Availability set** = spread VMs across fault/update domains.
- **Availability zone** = physically separate datacenter location in a region.
- **VM Scale Set** = many identical VMs that can autoscale.

Resources required for a VM

Common VM-related resource	Why it matters
Resource group	Management/lifecycle container
Virtual network and subnet	Private network placement
Network interface (NIC)	Network connectivity
Public IP (optional)	Internet-reachable access
Network security group (often used)	Traffic filtering rules
OS disk / data disks	Persistent storage
VM image	Starting operating system / template

What is tested: The exam may ask which supporting Azure resources are needed around a VM, especially networking and storage components.

Application hosting options

Hosting choice	Best fit	Why you choose it
Azure Web Apps	Standard web apps, APIs, fast deployment	Managed hosting for HTTP apps
Containers	Microservices, portable packaged apps	Consistent runtime across environments
Virtual Machines	Full custom environment or legacy app	Maximum control

Exam cue words

- “Do not manage servers” -> Web Apps or Functions.
- “Need custom OS settings” -> VMs.
- “Package once, run consistently” -> containers.

Virtual networking

Networking service / concept	What it does	Exam shorthand
Virtual Network (VNet)	Private network boundary in Azure	Azure’s logical network

Networking service / concept	What it does	Exam shorthand
Subnet	Segment within a VNet	Organize resources and apply controls
VNet peering	Connect VNets privately over Azure backbone	Fast private VNet-to-VNet connectivity
Azure DNS	Host and manage DNS domains / name resolution	DNS in Azure
VPN Gateway	Encrypted tunnel over the internet	Cheaper, internet-based hybrid link
ExpressRoute	Private dedicated connection to Azure	More predictable, private enterprise connectivity

Public endpoints vs private endpoints

Endpoint type	What it means	Best use case	Exam cue words
Public endpoint	Service reachable over public internet (secured by auth/firewalls)	Internet-facing apps or public APIs	public access, internet reachable
Private endpoint	Private IP in your VNet to reach an Azure service via Private Link	Private-only access to PaaS services	no public exposure, private IP, internal-only

Azure example: A storage account used by internal apps only can expose a private endpoint inside a VNet so traffic stays on private addressing.

Describe Azure storage services

What the exam tests

- Compare Azure Storage services
- Describe storage tiers
- Describe redundancy options
- Describe storage account options and types
- Moving files: AzCopy, Storage Explorer, File Sync
- Migration: Azure Migrate and Data Box

Compare Azure Storage services

Service	What it stores	Best fit	Example
Blob Storage	Object data	Images, backups, documents, data lakes	Store media uploads
Azure Files	Managed file shares	SMB/NFS shared file access	Lift-and-shift file share
Queue Storage	Messages	Decoupled app components	Background task queue
Table Storage	NoSQL key/attribute data	Simple large-scale structured data	Device metadata
Managed Disks	Block storage for VMs	VM OS and data disks	Persistent VM storage

Exam trick: Blob is not the same as Files. Blob is object storage; Files is a managed network file share.

Storage tiers

Tier	Access pattern	Cost pattern	Important exam note
Hot	Frequently accessed	Highest storage cost, lowest access cost	Default for active data
Cool	Infrequently accessed	Lower storage cost, higher access cost	Good for backup/older files
Cold	Rarely accessed but still online	Lower storage cost than Cool, higher retrieval cost	Online but optimized for very infrequent use
Archive	Very rarely accessed, offline tier	Lowest storage cost, highest retrieval delay/cost	Must rehydrate before use

Mnemonic: Hot = now, Cool = later, Cold = rare, Archive = deep storage.

Redundancy options

Option	Copies live where?	Best remembered as
LRS	Multiple copies in one datacenter	Cheapest basic redundancy
ZRS	Across availability zones in one region	In-region zone resilience
GRS	Primary region plus async secondary region	Cross-region resilience
RA-GRS	GRS plus read access to secondary	Read secondary during normal ops
GZRS	Zone redundancy in primary plus geo replication	Best combined regional + cross-region resilience
RA-GZRS	GZRS plus read access to secondary	Maximum resilience plus readable secondary

Exam tip: More resilience usually means more cost.

Storage account options and storage types

Storage account / performance option	Best fit
Standard general-purpose v2	Default choice for most Azure Storage workloads
Premium block blobs	High-performance object storage workloads
Premium file shares	High-performance Azure Files workloads
Premium page blobs / managed disk scenarios	High-performance disk-oriented workloads

Move files into and around Azure

Tool	Best use
AzCopy	Command-line, high-performance data copy to/from storage
Azure Storage Explorer	GUI for browsing and managing storage data
Azure File Sync	Cache Azure file shares on Windows Servers / hybrid file access

Migration options

Tool	Best use
Azure Migrate	Assess and migrate servers, databases, apps, and infrastructure workloads
Azure Data Box	Physical appliance/service for moving very large data sets when network transfer is too slow or impractical

Azure example: A company with petabytes of archived engineering data and limited network bandwidth may use Data Box.

Describe Azure identity, access, and security

What the exam tests

- Directory services: Microsoft Entra ID, Microsoft Entra Domain Services
- Authentication methods: SSO, MFA, passwordless
- External identities
- Microsoft Entra Conditional Access
- Azure RBAC
- Zero Trust
- Defense-in-depth model
- Microsoft Defender for Cloud

Microsoft Entra ID vs Microsoft Entra Domain Services

Service	Best description	Best fit
Microsoft Entra ID	Cloud-based identity and access management service	Users, groups, SSO, MFA, cloud app identity
Microsoft Entra Domain Services	Managed domain services in Azure	Legacy apps needing domain join, Kerberos, LDAP, or Group Policy without deploying domain controllers

Exam trick: Entra ID is not the same thing as classic on-prem AD DS. Entra Domain Services exists for legacy/domain-style needs in Azure.

Authentication methods

Method	What it means	Why it matters
Single sign-on (SSO)	One sign-in to access multiple apps	Better user experience, fewer repeated logins
Multifactor authentication (MFA)	Something you know + something you have/are	Stronger identity protection
Passwordless	Sign-in without traditional password	Reduced password risk and friction

External identities and Conditional Access

Topic	What to remember
External identities	Lets partners, guests, and external users collaborate securely
Conditional Access	If-then access policies based on signals such as user, location, device, or risk

Azure example: Require MFA only when a user signs in from outside trusted locations -> Conditional Access.

Azure RBAC

Azure role-based access control determines **who** can do **what** on **which scope**.

Element	Meaning
Security principal	User, group, service principal, or managed identity
Role definition	Set of allowed actions (Reader, Contributor, Owner, etc.)
Scope	Management group, subscription, resource group, or resource

Exam trick: RBAC grants permissions. It does not enforce configuration standards. That is Azure Policy.

Zero Trust and defense in depth

Zero Trust principles

- Verify explicitly
- Use least privilege
- Assume breach

Defense in depth layers

Physical -> Identity -> Perimeter -> Network -> Compute -> Application -> Data

What the exam tests

- Zero Trust is a security strategy.
- Defense in depth uses multiple layers so one control failure does not expose everything.

Microsoft Defender for Cloud

Microsoft Defender for Cloud helps improve security posture and provides security recommendations and workload protection across Azure, hybrid, and multicloud environments.

Best remembered as	Example exam interpretation
Cloud security posture + protection	“Which service gives security recommendations and posture visibility?”

Representative practice questions - Describe Azure architecture and services

Q1. Multiple choice

A company wants an application to continue running even if one datacenter in a region fails. Which Azure feature should it use?

- A. Management groups
- B. Availability zones
- C. Azure Policy
- D. Resource locks

Answer: B. Availability zones

Why: Availability zones are physically separate datacenter locations within a region and are designed for in-region

resiliency.

Skill tested: Describe the core architectural components of Azure - availability zones.

Q2. Multiple choice

You need to apply governance rules across several Azure subscriptions belonging to the same organization. What should you use?

- A. Resource groups
- B. Availability sets
- C. Management groups
- D. Azure DNS

Answer: C. Management groups

Why: Management groups sit above subscriptions and let you apply policies and access controls across multiple subscriptions.

Skill tested: Describe the core architectural components of Azure - management groups and hierarchy.

Q3. Scenario-based

A retail company needs hundreds of identical VMs to automatically increase during a sale and decrease afterward. Which option best fits?

- A. Availability sets
- B. Azure Virtual Desktop
- C. VM Scale Sets
- D. Azure Functions

Answer: C. VM Scale Sets

Why: VM Scale Sets are designed for large sets of identical VMs with scaling behavior.

Skill tested: Describe Azure compute and networking services - VM options.

Q4. Multiple choice

A company needs a private dedicated connection from its on-premises datacenter to Azure that does not traverse the public internet. Which service should it use?

- A. Azure DNS
- B. VNet peering
- C. VPN Gateway
- D. ExpressRoute

Answer: D. ExpressRoute

Why: ExpressRoute provides dedicated private connectivity to Azure; VPN Gateway uses the public internet.

Skill tested: Describe Azure compute and networking services - virtual networking.

Q5. Multiple choice

Which redundancy option provides zone redundancy in the primary region and also replicates to a secondary region, with read access to that secondary?

- A. LRS
- B. ZRS
- C. RA-GRS
- D. RA-GZRS

Answer: D. RA-GZRS

Why: RA-GZRS combines zone redundancy in the primary region, geo-replication to a secondary region, and read access to the secondary.

Skill tested: Describe Azure storage services - redundancy options.

Q6. Multiple choice

Which Azure service should you use to grant a user read-only access to a resource group?

- A. Azure Policy
- B. Conditional Access
- C. Azure RBAC
- D. Microsoft Defender for Cloud

Answer: C. Azure RBAC

Why: RBAC controls authorization by assigning built-in or custom roles at a scope such as a resource group.

Skill tested: Describe Azure identity, access, and security - Azure RBAC.

Describe Azure management and governance (30-35%)

Describe cost management in Azure

What the exam tests

- Factors affecting cost
- Pricing calculator
- Cost management capabilities
- Tags

Factors affecting costs

Cost factor	Why cost changes
Resource type	Different services cost differently
Consumption / usage	More CPU hours, storage, or transactions = more cost
Region	Pricing varies by region
Performance tier / SKU	Premium services cost more than standard/basic
Storage tier and redundancy	Hot or RA-GZRS generally costs more than cool or LRS
Network traffic	Egress/data transfer can affect cost
Commitment model	Reservations/commitments can reduce cost for steady workloads

Pricing calculator and cost management capabilities

Tool / feature	Purpose	Best exam clue
Azure Pricing Calculator	Estimate costs before deployment	“Estimate before you build”
Azure Cost Management	Analyze actual spend and trends	“Track and manage ongoing cost”
Budgets / alerts	Warn when spend approaches threshold	“Notify before overspend”
Cost analysis / exports	Break down costs by service, subscription, or tag	“Show where money goes”

Tags

Tags are name-value metadata pairs applied to Azure resources.

What tags do well	What tags do not do
Group costs by department/project/environment	They do not grant permissions
Improve reporting and organization	They do not prevent deletion
Support chargeback/showback	They do not enforce allowed configurations by themselves

Azure example: Tag resources with Dept=Finance, Env=Prod, and App=Payroll for cost reporting.

Describe features and tools for governance and compliance

What the exam tests

- Microsoft Purview
- Azure Policy
- Resource locks

Governance and compliance tools

Tool	Primary purpose	Best exam clue
Microsoft Purview	Data governance, catalog, compliance, and data estate visibility	discover/classify/govern data
Azure Policy	Enforce or audit configuration standards across resources	allow/deny/audit configuration
Resource locks	Prevent accidental deletion or modification	protect resource from change

Azure Policy vs RBAC vs Locks

Tool	Core question it answers
Azure Policy	“Is this resource compliant with standards?”
Azure RBAC	“Who is allowed to do this?”
Resource locks	“Can this resource be changed or deleted, even by authorized users?”

Resource lock types

Lock type	Effect
CanNotDelete	Authorized users can modify but not delete
ReadOnly	Prevents changes; resource effectively becomes read-only

Exam trick: Locks can block actions even if a user has RBAC permission.

Describe features and tools for managing and deploying Azure resources**What the exam tests**

- Azure portal
- Azure Cloud Shell (CLI and PowerShell)
- Azure Arc
- Infrastructure as code (IaC)
- Azure Resource Manager (ARM) and ARM templates

Management and deployment tools

Tool / concept	What it is	Best use
Azure portal	Browser-based GUI	Learn, administer, and configure resources visually
Azure Cloud Shell	Browser-accessible shell with Azure CLI or PowerShell	Quick administration from anywhere
Azure Arc	Extends Azure management to on-premises and multicloud resources	Central hybrid/multicloud control
Infrastructure as code	Define infra declaratively in files	Repeatable, versioned deployments
Azure Resource Manager (ARM)	Azure deployment/control plane	Deploy and manage Azure resources consistently
ARM templates	JSON-based IaC templates for ARM	Repeatable deployments

Cloud Shell quick facts

Fact	Why it matters on the exam
Supports Azure CLI and PowerShell	Expect questions that contrast both
Runs in browser	No local install needed
Authenticated to Azure	Fast admin experience
Good for quick commands	CLI/PowerShell from portal

IaC and ARM

What the exam wants:

- IaC = define infrastructure in code for repeatability and consistency.
- ARM = Azure's deployment and management layer.
- ARM templates = declarative JSON files used to deploy resources.

Azure example: Deploy the same resource group, storage account, and app service in test and production using the same template.

Describe monitoring tools in Azure

What the exam tests

- Azure Advisor
- Azure Service Health
- Azure Monitor (Log Analytics, alerts, Application Insights)

Monitoring tools comparison

Tool	Best purpose	Best exam clue
Azure Advisor	Personalized best-practice recommendations	improve cost, performance, reliability, security
Azure Service Health	Information about Azure service issues, maintenance, and advisories affecting you	service incident or planned maintenance
Azure Monitor	Platform for collecting/analyzing telemetry	central monitoring
Log Analytics	Query and analyze log data	KQL, log search
Alerts	Notify or trigger action from conditions	threshold reached, send notification
Application Insights	Application performance monitoring	request rates, failures, dependency tracking

Easy distinctions that show up often

If the question asks...	Best answer
"Which service recommends ways to optimize cost/performance/security?"	Azure Advisor
"Which service tells me about Azure outages or planned maintenance affecting my services?"	Azure Service Health
"Which service collects metrics and logs?"	Azure Monitor

If the question asks...	Best answer
“Which service helps analyze app response times and failures?”	Application Insights
“Which feature sends notifications when a metric crosses a threshold?”	Alerts

Mnemonic: Advisor advises, Monitor measures, Service Health informs, App Insights watches the app.

Representative practice questions - Describe Azure management and governance

Q1. Multiple choice

Before deploying a new solution, a team wants to estimate expected Azure costs. Which tool should it use?

- A. Azure Policy
- B. Azure Pricing Calculator
- C. Azure Service Health
- D. Azure Advisor

Answer: B. Azure Pricing Calculator

Why: The Pricing Calculator is used before deployment to estimate the cost of planned services.

Skill tested: Describe cost management in Azure - pricing calculator.

Q2. Multiple choice

A company wants to group Azure spending by department without changing permissions. Which feature should it use?

- A. Resource locks
- B. Tags
- C. Management groups
- D. Conditional Access

Answer: B. Tags

Why: Tags are metadata used for organization and cost reporting; they do not grant or deny access.

Skill tested: Describe cost management in Azure - tags.

Q3. Scenario-based

Your organization wants to ensure that only approved Azure regions can be used for new resources. Which tool is the best fit?

- A. Azure RBAC
- B. Azure Policy
- C. Azure Advisor
- D. Azure Monitor

Answer: B. Azure Policy

Why: Azure Policy can audit, deny, or remediate resource configurations against organizational standards such as allowed locations.

Skill tested: Describe features and tools for governance and compliance - Azure Policy.

Q4. Multiple choice

A critical production resource must not be deleted accidentally, but administrators may still need to edit it. Which should you apply?

- A. ReadOnly lock
- B. CanNotDelete lock
- C. Contributor role
- D. Management group

Answer: B. CanNotDelete lock

Why: CanNotDelete prevents deletion while still allowing authorized modifications. ReadOnly would block changes as well.

Skill tested: Describe features and tools for governance and compliance - resource locks.

Q5. Multiple choice

An operations team wants to manage on-premises servers and some multicloud resources through Azure as if they were Azure resources. Which service should it use?

- A. Azure Arc

- B. Azure DNS
- C. Azure Virtual Desktop
- D. Azure Files

Answer: A. Azure Arc

Why: Azure Arc extends Azure management and governance to servers, Kubernetes clusters, and other resources outside Azure.

Skill tested: Describe features and tools for managing and deploying Azure resources - Azure Arc.

Q6. Drag-and-drop style

Match each requirement to the best Azure monitoring tool.

Requirement	Correct match
Get recommendations to reduce cost and improve reliability	Azure Advisor
Be notified about an Azure outage affecting your subscription	Azure Service Health
Investigate application request failures and performance issues	Application Insights
Query collected logs across resources	Log Analytics

Why: AZ-900 often tests tool separation more than deep monitoring implementation.

Skill tested: Describe monitoring tools in Azure.

Full glossary of key terms and acronyms + most common exam tricks / gotchas

Glossary of key terms and acronyms

Term	Meaning
ARM	Azure Resource Manager, the deployment and management layer for Azure
Availability set	Spreads VMs across fault and update domains
Availability zone	Physically separate datacenter location inside an Azure region
Azure Arc	Extends Azure management to on-premises and multicloud resources
Azure Advisor	Recommendation engine for cost, security, reliability, performance, and operational excellence
Azure Monitor	Platform for metrics, logs, alerts, and telemetry analysis
Azure Policy	Enforces or audits organizational standards
Blob Storage	Object storage for unstructured data
CapEx	Capital expenditure; up-front spend
Conditional Access	If-then access policy engine based on signals such as location or risk
Data Box	Physical transfer option for very large data migrations
Entra Domain Services	Managed domain services for domain join, LDAP, Kerberos, etc.
ExpressRoute	Private dedicated connection from on-premises to Azure
GRS	Geo-redundant storage
GZRS	Geo-zone-redundant storage
Hybrid cloud	Combined on-prem/private and public cloud environment
IaaS	Infrastructure as a Service
IaC	Infrastructure as code
KQL	Kusto Query Language used in Log Analytics
LRS	Locally redundant storage
MFA	Multifactor authentication
Microsoft Entra ID	Azure's cloud identity and access management directory service
OpEx	Operating expenditure; ongoing usage-based spend
PaaS	Platform as a Service
Private endpoint	Private IP-based access to Azure PaaS through Private Link
Public endpoint	Internet-accessible service endpoint
RA-GRS	Read-access geo-redundant storage
RA-GZRS	Read-access geo-zone-redundant storage
RBAC	Role-based access control
Region pair	Cross-region relationship used for resiliency/recovery strategy
Resource group	Logical container for Azure resources
SaaS	Software as a Service
Service Health	Azure service issue and maintenance communication for your environment
SSO	Single sign-on

Term	Meaning
Subscription	Billing and management boundary in Azure
VNet	Azure Virtual Network
VPN Gateway	Encrypted connection to Azure over the internet
ZRS	Zone-redundant storage
Zero Trust	Security strategy built on verify explicitly, least privilege, and assume breach

Most common exam tricks / gotchas

1) Policy vs RBAC vs Locks

Tool	Remember this
Azure Policy	Controls compliance and allowed configurations
Azure RBAC	Controls permissions
Resource locks	Protect from accidental delete/change

2) Availability sets vs availability zones vs VM Scale Sets

Feature	Think
Availability set	Protect grouped VMs from host/update domain issues
Availability zone	Separate datacenter location in one region
VM Scale Set	Many identical VMs with scaling

3) Service Health vs Monitor vs Advisor

Tool	Think
Service Health	Azure has a problem or maintenance event
Azure Monitor	Observe metrics and logs
Azure Advisor	Improve what you already deployed

4) Blob vs Files

Storage service	Think
Blob Storage	Object storage
Azure Files	Shared file system access

5) VPN Gateway vs ExpressRoute

Service	Think
VPN Gateway	Encrypted tunnel over the public internet
ExpressRoute	Private dedicated enterprise connection

6) Public endpoint vs private endpoint

Endpoint	Think
Public endpoint	Reach over internet
Private endpoint	Reach with private IP in a VNet

7) Entra ID vs Entra Domain Services

Service	Think
Entra ID	Identity for cloud auth/access
Entra Domain Services	Managed domain features for legacy/domain-aware workloads

8) Cost tools

Need	Best answer
Estimate before deployment	Pricing Calculator
Analyze actual spend	Cost Management
Organize cost reports	Tags

9) Migration tools

Need	Best answer
Assess and migrate workloads	Azure Migrate
Ship huge data physically	Data Box

10) Hot/Cool/Cold/Archive

- Higher access frequency -> hotter tier.
- Lower storage cost usually means higher access/retrieval cost.
- Archive is offline and requires rehydration.

Final rapid-fire memory list

- Management groups govern subscriptions.
- Resource groups hold resources.
- Zones = datacenter resilience inside a region.
- Region pairs = cross-region recovery relationship.
- Functions = serverless.
- App Service = managed web hosting.
- RBAC = permissions.
- Policy = compliance.
- Locks = protection.
- Advisor = recommendations.
- Monitor = telemetry.
- Service Health = Azure incidents/maintenance.

Resources

Official Microsoft Learn study resources

- Certification page (AZ-900):
<https://learn.microsoft.com/en-us/credentials/certifications/azure-fundamentals/>
- Official study guide for Exam AZ-900:
<https://learn.microsoft.com/en-us/credentials/certifications/resources/study-guides/az-900>
- Free Microsoft practice assessment hub:
<https://learn.microsoft.com/en-us/credentials/certifications/practice-assessments-for-microsoft-certifications>
- Direct AZ-900 practice assessment page:
<https://learn.microsoft.com/en-us/credentials/certifications/azure-fundamentals/practice/assessment?assessment-type=practice&assessmentId=23&practice-assessment-type=certification>

Recommended official learning paths

- Describe cloud concepts:
<https://learn.microsoft.com/en-us/training/paths/microsoft-azure-fundamentals-describe-cloud-concepts/>
- Describe Azure architecture and services:
<https://learn.microsoft.com/en-us/training/paths/azure-fundamentals-describe-azure-architecture-services/>
- Describe Azure management and governance:
<https://learn.microsoft.com/en-us/training/paths/describe-azure-management-governance/>

Best way to use this packet

1. Read the exam overview and memorize the comparison tables.
2. Review each main skill area once without stopping.
3. Re-answer the practice questions from memory.
4. Revisit the glossary and gotchas the night before the exam.
5. Take the official Microsoft practice assessment and inspect every rationale, not just your score.

Closing note

AZ-900 rewards clean distinctions more than deep engineering detail. If you can reliably separate cloud concepts, service categories, connectivity options, storage redundancy choices, governance tools, and monitoring tools, you are in strong shape for the exam.